Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

Output ontology is HIGH-priority per elicitation Q3, which specifies that the deployment requires both a correct/incorrect verdict AND a substantive Hindi-language explanatory rationale. MILU's output ontology is a closed-form MCQ label space [Q33, Q55] — a single correct option index. The dataset analysis confirms empirically: every one of 245 reviewed examples has only question/option1–4/target fields with no rationale, explanation, or scoring dimension for free-text output [CRITICAL Concern 1]. Furthermore, MILU's documented pattern that models perform worst precisely in the deployment's priority domains — Arts & Humanities, Social Sciences, Law & Governance [Q6, Q20, Q62, Q71, Q77] — means the benchmark's accuracy ontology, even within its scope, gives weakest signal in the areas the deployment cares about most. This is a fundamental structural-validity violation.

ID	Evidence
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label (`target`). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

Strengths

- 41-subject hierarchical taxonomy enables domain-level performance analysis aligned with Indian exam structure [Q42]
- MCQ label correctness from real exam keys provides a credible objective floor [Q26]

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are MCQ option indices only [Q33, Q55] — they do not include rationale-quality categories required by deployment.

ID	Evidence
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label (`target`). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

OO-2: Missing categories: rationale correctness, rationale fluency in Hindi, pedagogical clarity, and explanation appropriateness — all required by elicitation Q3 but absent from MILU [CRITICAL Concern 1].

ID	Evidence

CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label (`target`). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.
WEB-14	2025 Hindi LLM benchmarking preprint notes Hindi instruction-following/conversational benchmarks are 'largely unavailable publicly'

OO-3: Within MCQ scope, no Western-value-encoded categories observed; primary issue is omission, not encoding bias.

OO-4: Stakeholder-driven taxonomy redesign is required: a rationale-quality rubric in Hindi must be added to evaluate the deployment's actual output.

ID	Evidence
WEB-14	2025 Hindi LLM benchmarking preprint notes Hindi instruction-following/conversational benchmarks are 'largely unavailable publicly'

OO-5: Documented harm: MILU's output ontology omits the entire rationale-quality dimension, violating structural validity for this deployment. Additionally, weakest performance in culturally relevant domains [Q6, Q71, Q77] further limits utility for History/Polity/GK signal.

ID	Evidence
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label (`target`). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

Requires Expert Verification

- Design of a Hindi rationale-quality rubric requires regional pedagogical expert input

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Remediation

Gap 1: No rationale-quality construct in MILU's output ontology, while deployment requires Hindi explanatory feedback [Q33, CRITICAL Concern 1, elicitation Q3]

Recommendation: Commission a rationale-evaluation rubric with Hindi-medium pedagogy experts covering factual accuracy, Manak Hindi fluency, pedagogical clarity, and code-mixing ceiling; use as a complementary evaluation alongside MILU's MCQ accuracy.